Indian Liver Patient Disorder

This project focuses on predicting outcomes using machine learning techniques. Feature selection methods such as RFE were applied to identify the most relevant features. Multiple models including Logistic Regression, SVM, Decision Tree, and Random Forest were trained and evaluated. Random Forest achieved the best performance with the highest accuracy.

Objective:

To develop a machine learning model for predicting disease outcomes using patient health data by applying feature selection techniques and comparing multiple classification algorithms to identify the most accurate model.

Dataset:

I used a healthcare dataset containing patient details like Age and biochemical parameters. It includes multiple features and a target variable indicating disease presence and it is used for **classification.**

No.of rows: 583

No.of columns: 11

No. of selected features: 3

These are the details of the biochemical parameters:

1. **TB (Total Bilirubin)**
   - Measures total bilirubin in blood
   - High values → possible liver dysfunction
2. **DB (Direct Bilirubin)**
   - Conjugated bilirubin processed by liver
   - Elevated → liver or bile duct issues
3. **Alkphos (Alkaline Phosphatase)**
   - Enzyme related to bile ducts
   - High levels → bile obstruction or liver disease
4. **Sgpt (ALT – Alanine Aminotransferase)**
   - Liver enzyme
   - High → liver cell damage

5. **Sgot (AST – Aspartate Aminotransferase)**
    - Enzyme found in liver & heart
    - High → liver damage (but less specific than ALT)
6. **TP (Total Protein)**
    - Total protein in blood (albumin + globulin)
    - Low → liver/kidney issues
7. **ALB (Albumin)**
    - Protein made by liver
    - Low → poor liver function
8. **A/G Ratio (Albumin/Globulin Ratio)**
    - Ratio of albumin to globulin
    - Helps assess liver & immune function
9. **Selector (Target Variable)**
    - Usually:
        - 1 → Liver disease
        - 2 → No liver disease

Methods:

- Data collection: Collected and understood the dataset.
- Preprocessing data: Performed data preprocessing (handling missing values and outliers).
- Data Analysis: Analyse the data by univariate and bivariate analysis. Visualized by tools like DataStudio and Tableau.
- Feature Selection: Applied feature selection using RFE to select important features.
- Training and Testing set: Trained multiple machine learning models such as Logistic Regression, SVM, Decision Tree, and Random Forest.
- Evaluation: Evaluated model performance using accuracy.
- Selected Model: Compared all models and selected the best-performing model.
- Deploy: Deploy the model by user input to predict the liver disorder.

Results:

```python
def final_result():
    loaded_model=pickle.load(open("Finalized model_ILPD.sav",'rb'))
    Age=int(input("Enter the Age:"))
    Alkphos=float(input("Enter the Alkphos:"))
    Sgot=int(input("Enter the Sgot:"))
    result=loaded_model.predict([[Age,Alkphos,Sgot]])
    return result
```

```python
final_result()
```

```
Enter the Age: 45
Enter the Alkphos: 65.23
Enter the Sgot: 89
```

```
array([1])
```

```python
final_result()
```

```
Enter the Age: 60
Enter the Alkphos: 481.75
Enter the Sgot: 34
```

```
array([0])
```

0-represent no disease.

1-represent have disease.

These are the summary of the project.